

Bohei-Shield.AI (BS.AI)

MAKAMI.AI - STANDALONE MULTI-AGENT HEALTH WRAPPER SPECIFICATION

SPEC v0.4

STANDALONE WRAPPER

FAIL-CLOSED ARCHITECTURE

NON-SLM

Executive Summary: Bohei-Shield.AI (BS.AI) is an independent, deterministic multi-agent health wrapper engineered for high-assurance environments. It operates strictly as a thermodynamic and policy validation boundary—it is neither a Small Language Model (SLM) nor a custom cryptographic cipher. Host systems pass a standardized HealthSnapshot, and the wrapper enforces structural integrity, hop caps, four-cache telemetry, the Hideo paranoia filter, and Minuteman dissipate-and-stand-down protocols without ever modifying or generating computational state.

1. Core Pipeline Architecture

Every message passing through the BS.AI wrapper traverses a strict 8-phase validation pipeline mapped from traditional martial arts principles to ensure fail-closed security:

Shin → Rei → Kumi-kata → Kuzushi → Tsukuri → Kake → Ukemi → Zanshin

Phase	Operational Function & Security Enforcement
Shin (Mind)	Core initialization and agent registry lookup. Establishes baseline execution context.
Rei (Etiquette)	Actor verification. Rejects unknown actors and enforces quarantine constraints for non-live agents.
Kumi-kata (Gripping)	Message typing validation. Refuses untyped or unlisted message structures, triggering immediate Minuteman dissipation (RT-4).
Kuzushi (Unbalancing)	Hop-count enforcement and Hideo filter evaluation. Limits lateral graph traversal and blocks circular overrides.
Tsukuri (Fitting)	Claim certification check. Validates marked payloads ([AXIOM], [SCHEMA], [BLUEPRINT], [LEAN]) against certified clerks.
Kake (Execution)	Action dispatch. Evaluates persist operations, quorum thresholds, and restores.
Ukemi (Receiving)	Thermodynamic stack health evaluation (κ , H). Freezes epoch and seals upon threshold breach.
Zanshin (Follow-through)	Telemetry recording, cryptographic digest chaining, and epoch advancement.

2. Locked Constants & System Parameters

Parameter	Locked Value
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Allowed Message Types	62
Active Lanes (LANES)	ingest, query, sibling, tool-result, persist, restore
Quorum Threshold (QUORUM)	32
Telemetry Caches	axiomatic, structural, dynamic, episodic
Default Hop Cap	8
Ukemi Kappa Threshold (κ)	0.85
Ukemi Energy Threshold (H)	0.90

3. Security Rules & Invariants

- **Minuteman Role Isolation (RT-10):** Minuteman is strictly a structural role triggered upon policy violation, never a persistent agent ID in the agent registry.
- **Fail-Closed Persistence (RT-7):** Plaintext persistence is strictly prohibited. Episodic caches require a sealed cryptographic envelope (`bsai_master.key`).
- **No Marked Mints (RT-8):** Tool-result messages are prohibited from minting marked claims and cannot originate from the operator ID.
- **Lane Integrity (RT-3):** Writing into closed or quarantined lanes is blocked unconditionally.

4. Telemetry & Cryptographic Chain

Every transaction appends an immutable audit record to the episodic and dynamic caches. Each record includes SHA-256 digest chaining (`prev_digest`) to ensure tamper-evident logging across epochs.

```
{
  "ts": 1723849200.12345,
  "cache": "episodic",
  "agent_id": "oracle",
  "lane": 7,
  "C": 0.74000,
  "kappa": 0.21000,
  "H": 0.33000,
  "key_epoch": 1,
  "live_faces": 62,
  "hops": 1,
  "decision": "ALLOW",
  "resource": "query",
  "prev_digest": "00000000000000000000000000000000",
  "digest": "a3f8bc91e2047d12... [truncated 32 chars]"
}
```

5. Python Implementation Reference

The core specification logic is fully self-contained within the reference module `BoheiShield`. It provides robust error handling, HKDF-derived envelope encryption via Python's `cryptography` library, and deterministic simulation execution via its built-in `demo()` harness.